

Package ‘shinyHierarchy’

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Title Hierarchical Input Component for Shiny

Version 0.1.0

Description Hierarchical slicer-style input for Shiny applications with multi-selection, tri-state checkboxes, search, cascading selection, and a structured selection value for analytics workflows.

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Encoding UTF-8

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Imports shiny, htmltools, jsonlite

Suggests testthat (>= 3.0.0)

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buildHierarchyTree	<i>Build a hierarchical tree from tabular data</i>
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Description

Converts a data frame and an ordered set of columns into a nested hierarchy suitable for the Shiny input component.

Usage

```
buildHierarchyTree(data, levels)
```

Arguments

- data A data frame containing the hierarchy columns.
- levels A character vector with column names, ordered from highest to lowest hierarchy level.

Value

A list representing the hierarchy tree.

Examples

```
data <- data.frame(
  year = c(2026, 2026, 2026),
  month = c("January", "January", "February"),
  day = c(1, 2, 1)
)
buildHierarchyTree(data, c("year", "month", "day"))
```

hierarchyInput	<i>Hierarchical tree input for Shiny</i>
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Description

Creates a hierarchical slicer-style input similar to a Power BI date hierarchy. The input value is a structured object (schema version 1) with resolved, rollup, explicit, and summary fields.

Usage

```

hierarchyInput(
  inputId,
  label = NULL,
  data,
  levels,
  selected = NULL,
  expanded = NULL,
  placeholder = "Selección múltiple",
  display = c("inline", "dropdown"),
  search = FALSE,
  searchPlaceholder = "Buscar...",
  selectAll = TRUE,
  selectAllLabel = "Seleccionar todo",
  expand = c("first", "all", "none"),
  width = "100%",
  height = "360px"
)

```

Arguments

inputId	The input slot that will be used to access the value.
label	Optional label displayed above the component.
data	A data frame containing the hierarchy columns.
levels	Character vector with column names, ordered from highest to lowest hierarchy level.
selected	Optional character vector of node ids to select initially.
expanded	Optional character vector of node ids to expand initially.
placeholder	Text shown when nothing is selected.
display	Visual presentation: "inline" keeps the tree visible, while "dropdown" opens it in a floating panel.
search	Whether to show a search box above the tree.
searchPlaceholder	Placeholder for the search field.
selectAll	Whether to show the select-all row.
selectAllLabel	Label for the select-all control.
expand	Initial expansion mode: first level, all levels, or none.
width	CSS width of the component.
height	CSS max-height of the tree area.

Details

The value received from the browser is user-controlled and must not be used as an authorization mechanism. Applications that filter sensitive data must validate selected ids against the server-side hierarchy and permissions.

Value

A Shiny HTML object.

hierarchyPath	<i>Extract path values at a given hierarchy level</i>
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Description

Extract path values at a given hierarchy level

Usage

```
hierarchyPath(value, level_name)
```

Arguments

value	Value returned by a hierarchy input (input\$...).
level_name	Name of the hierarchy level column.

Value

A character vector of unique values at that level among resolved selections.

hierarchyResolved	<i>Extract resolved leaf selections from a hierarchy input value</i>
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Description

Extract resolved leaf selections from a hierarchy input value

Usage

```
hierarchyResolved(value)
```

Arguments

value	Value returned by a hierarchy input (input\$...).
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Value

A list of resolved node records, or an empty list.

hierarchyRollup	<i>Extract rollup selections from a hierarchy input value</i>
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Description

Extract rollup selections from a hierarchy input value

Usage

```
hierarchyRollup(value)
```

Arguments

value	Value returned by a hierarchy input (input\$...).
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Value

A list of rollup node records, or an empty list.

updateHierarchyInput	<i>Update a hierarchical input from the server</i>
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Description

Update a hierarchical input from the server

Usage

```
updateHierarchyInput(  
  session = shiny::getDefaultReactiveDomain(),  
  inputId,  
  data = NULL,  
  levels = NULL,  
  selected = NULL,  
  expanded = NULL,  
  clear = FALSE,  
  open = NULL  
)
```

Arguments

<code>session</code>	Shiny session object.
<code>inputId</code>	The input id to update.
<code>data</code>	Optional new data frame.
<code>levels</code>	Optional hierarchy column names (required if data is set).
<code>selected</code>	Optional vector of node ids to select (cascades to descendants).
<code>expanded</code>	Optional vector of node ids to expand.
<code>clear</code>	If TRUE, clears the current selection.
<code>open</code>	If TRUE or FALSE, opens or closes the dropdown panel.

Value

No useful return value, invisibly NULL.

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